

Stat 341 – Project Milestone 2

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```
mlvar_data = read.csv("SA_data_for_mlvar_ana.csv")
#make subjects and dyads factors
mlvar_data$subject <- as.factor(mlvar_data$subject)
mlvar_data$dyad <- as.factor(mlvar_data$dyad)
#compute synchrony, log transform to correct for heteroscedasticity
mlvar_data$Synchrony <- -log(mlvar_data$dtw)
colnames(mlvar_data) <- c('X', 'dyad', 'subject', 'dtw', 'EyeContact', 'PupilSize', 'Synchrony')

glimpse(mlvar_data)
```

```
## Rows: 56,400
## Columns: 7
## $ X          <int> 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 1~
## $ dyad       <fct> D001, D001, D001, D001, D001, D001, D001, D001, D001, D001,~
## $ subject    <fct> S028, S028, S028, S028, S028, S028, S028, S028, S028, S028,~
## $ dtw        <dbl> 26.038945, 42.312825, 27.318998, 37.486558, 15.107104, 21.0~
## $ EyeContact <dbl> 0.000, 0.000, 0.033, 0.256, 0.350, 0.000, 0.000, 0.085, 1.0~
## $ PupilSize  <dbl> 0.35092227, 0.47939718, 0.47493465, 0.55883216, 0.37438550,~
## $ Synchrony  <dbl> -3.2595933, -3.7450902, -3.3075824, -3.6239824, -2.7151651,~
```

The data is from a research conducted by Sophie Wohltjen and Thalia Wheatley. Their research paper can be found here: <https://www.pnas.org/doi/10.1073/pnas.2106645118>

This research paper examines the role of eye contact in conversation. The authors found that when two people are talking, eye contact is a key factor in determining when shared attention between them is rising or falling. It was found that when two people are engaging in conversation, eye contact increases as the conversation progresses, indicating a rise in shared attention.

For this project, the question we are interested in answering is: How does eye contact affect shared attention in conversations?

Our predictor variables are eye contact and pupil size the response variable is shared attention in conversation (synchrony). Our predictor variables are eye contact and pupil size. Our response variable is shared attention in conversation (synchrony).

Causal Diagram

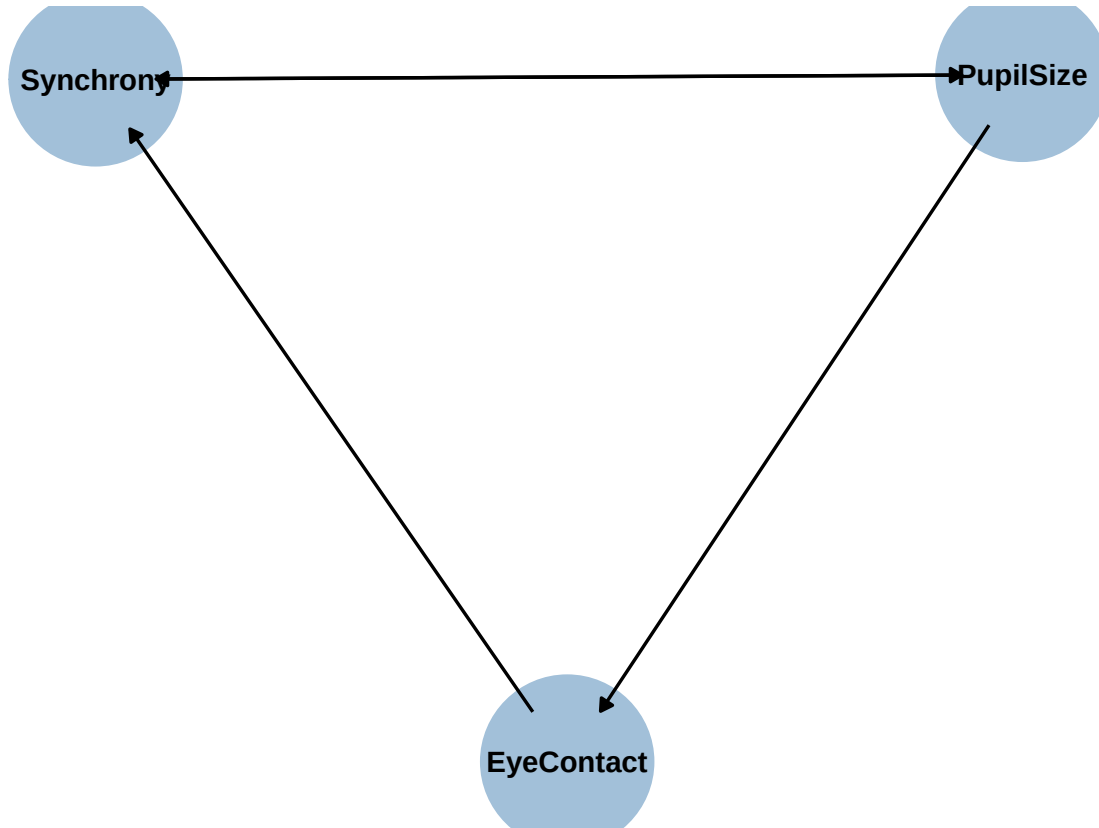
```
library(dagitty)
library(CalvinBayes)

precision_ex <- dagitty("dag{
PupilSize -> Synchrony
Synchrony -> PupilSize
PupilSize -> EyeContact
EyeContact -> Synchrony
```

```

}))
gg_dag(precision_ex,
      size = 30,
      highlight = 'A')

```



Model Description

$$\text{synchrony}_i \sim \text{Normal}(\mu_i, \sigma)$$

$$\mu_i = \beta_0 + \beta_1 * \text{EyeContact}_i + \beta_2 * \text{PupilSize}_i$$

$$\beta_0 \sim \text{Normal}(2, 0.5)$$

$$\beta_1 \sim \text{Normal}(0.1, 0.5)$$

$$\beta_2 \sim \text{Normal}(0.02, 0.05)$$

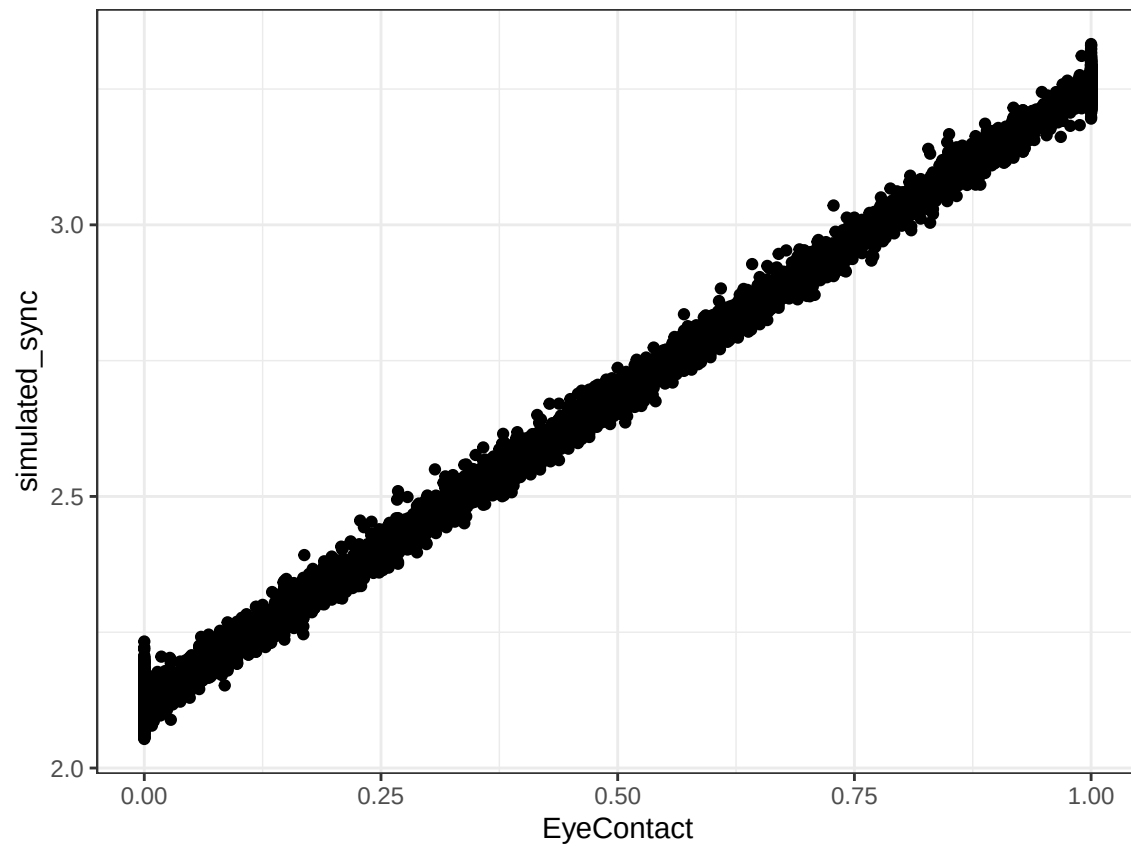
$$\sigma \sim \text{Normal}(\text{mean}_2 = 1, \text{sd}_2 = 1)$$

Prior Predictive Check

```
b0 = rnorm(1, mean = 2, sd = 0.5)
b1 = rnorm(1, mean = 0.1, sd = 0.5)
b2 = rnorm(1, mean = 0.02, sd = 0.05)

simulated_sync <- b0 + b1 * mlvar_data$EyeContact + b2 * mlvar_data$PupilSize

gf_point(simulated_sync ~ EyeContact, data = mlvar_data)
```



```
gf_point(simulated_sync ~ PupilSize, data = mlvar_data)
```

